

Algoritmos de Assinatura		Algoritmos de Troca de Chaves																	Latência do Handshake [ms] — Menor é Melhor
		P256 + HQC128	HQC128	X25519 + HQC128	P256 + FRODO640SHAKE	P256 + FRODO640AES	BP256 + MLKEM512	P256 + BIKEL1	P256 + MLKEM512	FRODO640SHAKE	X25519 + FRODO640SHAKE	FRODO640AES	X25519 + FRODO640AES	BIKEL1	X25519 + BIKEL1	X25519 + MLKEM512	MLKEM512		
Algoritmos de Assinatura	MLDSA44	36.1 ±0.4	32.0 ±0.2	32.2 ±0.4	10.0 ±0.3	8.1 ±0.3	8.1 ±0.3	7.9 ±0.4	7.1 ±0.3	6.0 ±0.3	6.4 ±0.5	4.1 ±0.2	4.4 ±0.3	4.0 ±0.5	4.0 ±0.3	3.4 ±0.6	2.7 ±0.8		
	FALCONPADDED512	36.1 ±0.1	32.1 ±0.1	32.3 ±0.1	10.2 ±0.0	8.3 ±0.1	8.1 ±0.1	7.9 ±0.0	7.0 ±0.1	6.2 ±0.0	6.4 ±0.1	4.3 ±0.1	4.5 ±0.1	4.0 ±0.1	4.2 ±0.1	3.3 ±0.1	2.5 ±0.1		
	FALCON512	36.0 ±0.1	32.1 ±0.1	32.3 ±0.1	10.2 ±0.1	8.3 ±0.1	8.2 ±0.1	7.9 ±0.0	7.1 ±0.1	6.3 ±0.1	6.5 ±0.1	4.3 ±0.1	4.5 ±0.1	4.0 ±0.1	4.2 ±0.1	3.3 ±0.1	2.5 ±0.1		
	P256 + MLDSA44	36.3 ±0.1	32.9 ±0.1	32.6 ±0.1	10.3 ±0.1	8.4 ±0.1	8.6 ±0.1	8.2 ±0.1	7.3 ±0.1	7.0 ±0.0	6.8 ±0.1	5.1 ±0.1	4.8 ±0.1	4.8 ±0.0	4.6 ±0.1	3.6 ±0.1	3.3 ±0.1		
	P256 + FALCONPADDED512	36.4 ±0.2	33.0 ±0.1	32.8 ±0.1	10.5 ±0.1	8.5 ±0.1	8.7 ±0.0	8.3 ±0.1	7.5 ±0.1	7.1 ±0.0	6.9 ±0.1	5.2 ±0.1	5.0 ±0.1	4.9 ±0.1	4.6 ±0.1	3.8 ±0.1	3.5 ±0.1		
	P256 + FALCON512	36.5 ±0.1	33.0 ±0.1	32.8 ±0.1	10.4 ±0.1	8.6 ±0.1	8.7 ±0.1	8.3 ±0.1	7.5 ±0.1	7.1 ±0.1	6.9 ±0.1	5.2 ±0.1	5.0 ±0.1	4.9 ±0.1	4.6 ±0.1	3.8 ±0.0	3.5 ±0.1		
	RSA3072 + MLDSA44	38.1 ±0.1	34.6 ±0.1	34.3 ±0.1	12.2 ±0.1	10.3 ±0.1	10.4 ±0.1	10.0 ±0.0	9.2 ±0.1	8.6 ±0.1	8.4 ±0.1	6.8 ±0.1	6.5 ±0.1	6.5 ±0.1	6.2 ±0.1	5.3 ±0.1	5.1 ±0.1		
	RSA3072 + FALCONPADDED512	38.3 ±0.2	34.7 ±0.1	34.4 ±0.1	12.3 ±0.1	10.4 ±0.1	10.4 ±0.1	10.1 ±0.1	9.3 ±0.1	8.8 ±0.1	8.5 ±0.1	6.9 ±0.1	6.5 ±0.0	6.5 ±0.0	6.3 ±0.1	5.5 ±0.0	5.3 ±0.1		
	RSA3072 + FALCON512	38.3 ±0.1	34.7 ±0.1	34.4 ±0.1	12.3 ±0.1	10.4 ±0.1	10.5 ±0.1	10.1 ±0.1	9.3 ±0.1	8.8 ±0.1	8.6 ±0.1	6.8 ±0.1	6.6 ±0.1	6.6 ±0.1	6.3 ±0.1	5.5 ±0.1	5.3 ±0.1		
	SPHINCSSHA2128FSIMPLE	43.7 ±0.1	39.7 ±0.1	39.8 ±0.1	17.7 ±0.1	15.8 ±0.1	15.9 ±0.1	15.7 ±0.1	14.7 ±0.0	13.7 ±0.1	14.0 ±0.1	11.9 ±0.1	12.1 ±0.1	11.6 ±0.1	11.8 ±0.1	10.9 ±0.1	10.3 ±0.1		
	P256 + SPHINCSSHA2128FSIMPLE	44.1 ±0.1	40.6 ±0.1	40.4 ±0.1	18.0 ±0.1	16.2 ±0.1	16.2 ±0.1	15.9 ±0.1	15.1 ±0.1	14.7 ±0.1	14.4 ±0.1	12.8 ±0.2	12.5 ±0.1	12.5 ±0.1	12.3 ±0.1	11.4 ±0.1	11.2 ±0.1		
	RSA3072 + SPHINCSSHA2128FSIMPLE	45.8 ±0.1	42.3 ±0.1	42.0 ±0.1	19.9 ±0.1	18.0 ±0.1	18.1 ±0.1	17.7 ±0.1	16.8 ±0.1	16.3 ±0.1	16.1 ±0.1	14.5 ±0.1	14.2 ±0.1	14.2 ±0.1	13.9 ±0.1	13.1 ±0.1	12.8 ±0.1		
	SPHINCSSHAKE128FSIMPLE	52.4 ±0.1	48.4 ±0.1	48.6 ±0.1	26.3 ±0.1	24.5 ±0.1	24.6 ±0.0	24.3 ±0.1	23.5 ±0.1	22.4 ±0.1	22.7 ±0.1	20.5 ±0.1	20.7 ±0.1	20.5 ±0.1	20.5 ±0.1	19.6 ±0.1	19.0 ±0.1		
	P256 + SPHINCSSHAKE128FSIMPLE	52.8 ±0.0	49.3 ±0.1	49.1 ±0.1	26.8 ±0.1	24.8 ±0.2	25.0 ±0.1	24.6 ±0.0	23.8 ±0.1	23.3 ±0.1	23.1 ±0.1	21.5 ±0.1	21.2 ±0.1	21.2 ±0.1	21.0 ±0.1	20.1 ±0.1	19.8 ±0.1		
	RSA3072 + SPHINCSSHAKE128FSIMPLE	54.5 ±0.2	50.9 ±0.1	50.7 ±0.1	28.6 ±0.1	26.7 ±0.1	26.7 ±0.1	26.5 ±0.1	25.6 ±0.1	25.0 ±0.1	24.8 ±0.2	23.1 ±0.1	22.9 ±0.1	22.8 ±0.1	22.6 ±0.1	21.7 ±0.1	21.5 ±0.1		
	SPHINCSSHA2128SSIMPLE	182.4 ±0.1	178.5 ±0.1	178.6 ±0.1	156.7 ±0.2	154.9 ±0.2	154.6 ±0.1	154.2 ±0.1	153.6 ±0.1	152.7 ±0.2	153.2 ±0.2	150.9 ±0.2	151.2 ±0.1	150.4 ±0.1	150.6 ±0.1	149.8 ±0.1	149.1 ±0.1		
	P256 + SPHINCSSHA2128SSIMPLE	182.8 ±0.2	179.4 ±0.1	179.2 ±0.2	157.0 ±0.3	155.3 ±0.2	155.0 ±0.1	154.7 ±0.1	154.2 ±0.2	153.7 ±0.2	153.4 ±0.1	151.8 ±0.2	151.7 ±0.1	151.3 ±0.1	151.2 ±0.1	150.4 ±0.2	150.0 ±0.2		
	RSA3072 + SPHINCSSHA2128SSIMPLE	184.6 ±0.1	181.1 ±0.1	180.8 ±0.1	158.7 ±0.1	157.0 ±0.2	156.8 ±0.1	156.6 ±0.1	156.1 ±0.2	155.2 ±0.2	155.1 ±0.2	153.3 ±0.2	153.5 ±0.2	152.9 ±0.1	152.8 ±0.1	152.0 ±0.2	151.8 ±0.2		
SPHINCSSHAKE128SSIMPLE	351.1 ±0.2	347.1 ±0.1	347.6 ±0.2	325.6 ±0.2	323.8 ±0.2	323.6 ±0.1	323.2 ±0.1	322.8 ±0.3	321.6 ±0.2	321.4 ±0.2	320.0 ±0.2	319.8 ±0.2	319.3 ±0.1	319.5 ±0.2	318.9 ±0.1	318.1 ±0.3			
P256 + SPHINCSSHAKE128SSIMPLE	351.6 ±0.1	348.3 ±0.2	347.8 ±0.2	326.1 ±0.3	323.9 ±0.5	323.7 ±0.2	323.6 ±0.1	322.9 ±0.1	322.9 ±0.3	322.5 ±0.2	320.6 ±0.3	320.6 ±0.2	320.4 ±0.2	319.9 ±0.2	319.3 ±0.2	319.2 ±0.2			
RSA3072 + SPHINCSSHAKE128SSIMPLE	353.7 ±0.2	349.7 ±0.1	349.5 ±0.2	328.7 ±0.3	325.8 ±0.1	325.5 ±0.1	325.6 ±0.1	324.9 ±0.2	324.2 ±0.2	324.1 ±0.2	322.6 ±0.3	322.3 ±0.3	322.0 ±0.2	321.5 ±0.1	320.8 ±0.1	320.8 ±0.2			